Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/26/2006
PI Name:	Landau, David Paul Ph.D.		
Project Title:	Computer Simulations of Confined Quantum Systems in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	30602	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	04/01/2004	**End Date:**	12/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 12/31/2007 per PI's institution and GRC document (10/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	GSN 4515		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		

Task Description:	The objective is to use computer simulations of appropriate models to improve our understanding of the behavior of quantum systems in confined geometries as the phase transition is approached. Our immediate objective is to use the XY-model of "spins" to describe the critical behavior of superfluid helium. This is accomplished by an appropriate renaming of variables that yields a physically relevant description of the lambda point. We are developing improved methods for carrying out Monte Carlo simulations for systems with two parallel surfaces that play the role of the confining walls. Then, new spin dynamics methods will be used to study the time dependent behavior of the model and thus the thermal conductivity. Data will be compared with those from the BEST and DYNAMX experiments. Path integral Monte Carlo methods are also to be developed to study the behavior of confined Boson systems in different types of fields. Both the techniques developed and the results appropriate to helium may play an important role in the elucidation of behavior of other confined quantum systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	A finite size scaling analysis has been carried out for thermal conductivities of pseudo-two dimensional systems of liquid Helium as determined by computer simulations. For the range of aspect rations and temperatures that could be considered, the overall quality of the scaling was excellent with the same value of the critical exponent as for pore geometry. This calls into question the viability of the simple exponent prediction from the theory of dynamic scaling. Extensive simulations, using path integral Quantum Monte Carlo methods, have also now been completed for weakly interacting hard-core Bosons in trapping fields. Using an anisotropic pseudopotential for long range dipolar interactions and a hard sphere potential for the short range s-wave interaction, we determined the energetics and structural properties as a function of temperature and number of particles. For a cigar shaped trap the net contribution of the dipolar interactions is attractive and the density profiles shrink. For a disk shaped trap the net effect of the dipolar forces is repulsive and the density profiles expand. Lastly, we have also used path integral Monte Carlo simulations to examine dilute, trapped Bose gases in a quasi-two-dimensional geometry. For repulsive interactions the minimum value of the aspect ratio for which the system starts to behave like a quasi-two-dimensional Bose gas increases as the two-body interactions strength increases. The superfluid fraction for a quasi-two-dimension Bose gas is distinctly different that for either a quasi-one-dimensional Bose case or for a true three dimensional system, i.e. it decreases faster than for either of the other cases as the temperature increases and shows a stronger dependence on the interactions strength. It also lies well below the predictions from a semi-classical approximation.
Bibliography Type:	Description: (Last Updated: 02/26/2014)
Articles in Peer-reviewed Journals	Nho K, Landau DP. "Bose-Einstein condensation of trapped atoms with dipole interactions." Phys Rev A. 2005 Aug;72(2):023615. http://dx.doi.org/10.1103/PhysRevA.72.023615 , Aug-2005